

BRAC UNIVERSITY

Department of Computer Science and Engineering

Examination : Semester Midterm
Duration: **1 Hour 20 Minutes**

Semester: Fall 2025
Full Marks: **50**

CSE421 / EEE465 : Computer Networks

Answer **ALL** questions. (Pages: 2)

Figures in the right margin indicate marks.

Name:	ID:	Section:
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Q1 [CO1]	A software bug causes encrypted data to be delivered directly to the application layer without establishing a session. Identify the OSI layers that have been bypassed or mis-sequenced, and describe the problems Bob would observe at the application side. <i>Session. ② ②</i>	4
Q2 [CO2]	A news website stores all article metadata under <code>/articles/{id}/info.json</code> on its origin server. A university network uses a proxy cache to speed up web access for students. At 2:00 PM, the content team updates <code>/articles/55/info.json</code> directly on the origin server. The proxy still contains an older cached version from before the update. At 2:06 PM, a student submits a comment on the same article using a form that sends data to <code>/articles/55/comments</code>. <i>② POST</i> I. State which HTTP method the browser will use to submit this comment. Briefly explain. At 2:07 PM, another student requests: <code>GET /articles/55/info.json</code> through the proxy. II. State which version of the file the proxy will return to the student, and why? <i>③ Updated one. Conditional GET / HEADER if modified.</i>	2 + 4
Q3 [CO2]	FastFoodHub is an online food ordering website. When customers add items to their cart and refresh the page, the cart items stay saved. But some customers complain that on another day, when they use a different device, their cart becomes empty even though they are logged into the same account. <i>① Cookies</i> I. What web technology is being used to keep the cart items saved on the same device? <i>Diff Browser</i> II. Why do the cart items not appear when the customer switches to a different device? <i>Diff device</i>	2 + 2
Q4 [CO2]	Jannat tries to open <code>research.university.edu</code>, but her browser keeps showing an error that the site cannot be reached. Her teammate, who is using a different network, can open the same website instantly. Later, Jannat clears her DNS cache, and after that, the website loads normally on her device. <i>① URL → IP</i> I. Briefly explain how DNS resolution enables access to a website, and the reason why clearing the DNS cache solved Jannat's problem. <i>DNS → Host ②</i> II. Discuss why recursive resolution is convenient for clients while iterative resolution benefits DNS infrastructure. <i>Recursive ①.5 - Iterative ①.5</i>	3 + 3
Q5 [CO2]	Anika is attending an online lecture with 120 participants while also downloading a 200 MB course file in the background. During the lecture, a few words drop out because of weak Wi-Fi, but the session continues without stopping. The file download takes longer but finishes perfectly	4 + 2

with no missing data.

- I. If the online lecture used the same protocol as the file download, describe what would happen to the live session and explain why this would be a problem.
- II. Explain why the file took longer time to download. *Bandwidth division / TCP*

Q6 Consider a TCP connection using slow start and congestion avoidance.

- The initial slow-start threshold (ssthresh) is 15 MSS.
- The congestion window (cwnd) changes over the rounds as shown:

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13
cwnd (MSS)	1	2	4	8	15	16	17	18→9	10	11	?	13→1	2	4

- I. Identify why **cwnd = 15** at round 4. *cwnd = ssthresh*
- II. What will be the **cwnd** at round 10? *12 (CA)*
- III. Between **Round 7 and 11**, which one indicates more severe congestion? *11. RTO.*

Q7 Asif loads an HTML page in Google Chrome over a **persistent HTTP connection** (TCP termination time and the third phase of handshaking are negligible). The page contains **11 objects** in total, including the base HTML file. The size of each object is **4 MB**. The **total round-trip time** to load the page is **455 ms**. Note that the **TCP RTT** is twice the **per-object HTTP request-and-response RTT**. The server's bandwidth is **16 Mbps**.

I. Calculate the **TCP RTT** and the **per-object HTTP request response RTT**.

II. Calculate the **total page load time**. *$T = \frac{4 \times 8 \times 11}{16} \times 1000 + 455 = 2245 \text{ ms}$*

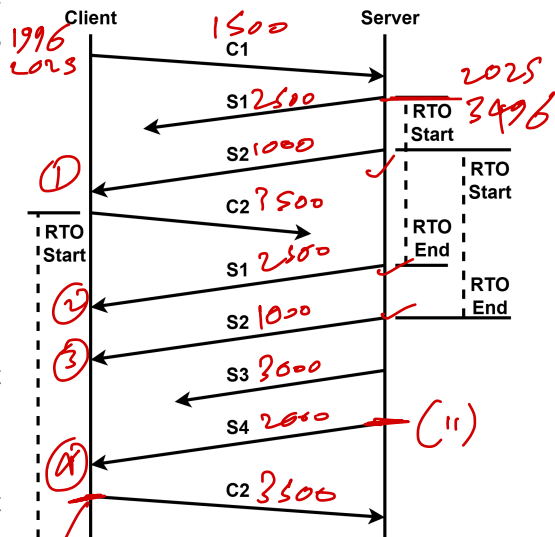
III. If the page had 20 objects instead of 11, explain which part of the delay would increase the most in persistent HTTP, and why. *RTT in seconds / RTT HTTP*

Q8 At a certain point, the client transmits segment **C1** with a sequence number of **1996** and an acknowledgment number of **2025**. The client's **rwnd** is **12000** bytes, and the server's **rwnd** is **10000** bytes before segment **C1** was sent. The data size of the **C1**, **C2**, **S1**, **S2**, **S3**, **S4** are **1500**, **3500**, **2500**, **1000**, **3000**, and **2000** bytes, respectively.

I. Determine whether Selective Repeat or Go-Back-N is used. Note that when the client sends the **second C2 segment**, the advertised **rwnd** is **8500** bytes.

II. Find the sequence and acknowledgment numbers of the **S4** segment.

III. Explain why the client retransmitted the **C2** segment, even though the **RTO** for **C2** did not expire.



(I) $r_{c2} = 8500 = 12000 - 2500 - 1000 = 8500$ (Go-Back-N) (S4 not stored)

(II) $S = 2025 + 2500 + 1000 + 3000 = 8525$

(III) $a = 3496$ (11) 3 Dup Ack

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Q1 [CO1]	A host can successfully communicate with another device on the same LAN (ARP reply received, frames exchanged), but cannot reach a web server on a different network. I. Which OSI addressing level is confirmed to be working? <i>Data Link.</i> II. Which OSI addressing level is likely misconfigured or failing? <i>Network.</i>	4
Q2 [CO2]	A university uses a web proxy cache to serve frequently accessed web objects. At 10:58, the content on the origin server is updated by the content team outside the local network, specifically, a new version of /banner.jpg is uploaded directly to the origin server. The proxy inside the university still holds an older cached copy from before 10:58. - At 10:59, a student attempts to log in by sending POST /login with JSON credentials through the proxy. - At 11:00, another student requests GET /banner.jpg via the proxy. I. When the proxy receives POST /login, should it return a cached response or forward the request to the origin server? Explain briefly <i>Forward. Send info.</i> II. Because the origin server's /banner.jpg was updated externally at 10:58, what mechanism should the proxy use to avoid serving the stale cached image to users? Briefly explain. <i>Conditional GET. / HEADER if modified.</i>	3 + 3
Q3 [CO2]	Sumya, a lecturer, sends assignment guidelines to 50 students by email. Each email goes from her laptop to the university's outgoing mail server, and then to the mail servers of the students. Later, 10 students, trying to open the email, notice that the email disappears from the server after they download it to their devices. I. Identify the two main email protocols used in this process. <i>SMTP + POP/POP3</i> II. Explain why both protocols are needed for complete email communication. Also, describe what would happen if only one of these protocols were used. <i>Send + Receive.</i>	2 + 2
Q4 [CO2]	Arman tries to visit library.campus.net, but his browser returns an error saying "server not found". He then manually tries the server's IP address and successfully loads the website. I. What does this tell you about where the failure occurred in the name-to-access process, and how does DNS normally map a hostname to an IP address? <i>DNS Misconfiguration.</i> II. Later, Arman inspects the DNS response and notices that the lookup for campus.net returns a hostname related to mail.campus.net. State what type of DNS record he received, and which type of record he should have received to load the website.	3 + 3

MX, A
②, ①

(Please Turn Over) 1

$$4 + 2$$

- I.** If the email sending were to use the same protocol as the live streaming, describe how this would affect the email transmission, and explain why it would be problematic.
- II.** Some of Rehman's friends notice that the live stream starts lagging while he is uploading the assignment. Briefly explain why uploading a large file can cause the live IPTV stream to lag.

Q6
[CO3]

- The initial slow-start threshold (ssthresh) is 6 MSS.
- The congestion window (cwnd) changes over the rounds as shown:

Round	0	1	2	3	4	5	6	7	8	9	10	11	12	13
cwnd (MSS)	1	2	4	6	7	8	9 → ?	2	4	5	6	7	8	9

- I.** Identify why **cwnd** = 6 at round 3. *ssthresh*
- II.** What will be the **cwnd** at the end of round 6? *1*
- III.** Between **Round 4 and 6**, which one indicates more severe congestion? *6, RTO*

Q7

- [CO3] I. Calculate the **TCP RTT** and the **per-object HTTP request response RTT**. 13.7
- [CO3] II. Calculate the **total page-load time**. $T = 1280 + \frac{24 \times 16}{0.32} \times 1000 = 13280 / 13.7$
- [CO2] III. Suppose the RTT becomes larger due to network congestion. Explain which part of the page-load time will increase, and briefly explain why. 1280 $(7) + (1)$

Q8

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|-------|------|---|
| [CO3] | I. | Determine whether Selective Repeat or Go-Back-N is used. Note that when the client sends the second C2 segment , the advertised rwnd is 7500 bytes. |
| [CO3] | II. | Find the sequence and acknowledgment numbers of the S3 segment. |
| [CO2] | III. | Explain why the server retransmitted both S1 & S2 segments, although S2 reached the client. |

